

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

45000126 - Nanotechnology

DEGREE PROGRAMME

04MI - Grado En Ingenieria De Materiales

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

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1. Description

1.1. Subject details

Name of the subject	45000126 - Nanotechnology
No of credits	6 ECTS
Type	Compulsory
Academic year of the programme	Third year
Semester of tuition	Semester 6
Tuition period	February-June
Tuition languages	English
Degree programme	04MI - Grado en Ingeniería de Materiales
Centre	04 - E.T.S. De Ing. De Caminos Canales Y P.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Jorge Pedros Ayala	B-308	j.pedros@upm.es	Th - 11:00 - 12:00
Fernando Calle Gomez (Subject coordinator)	Aula 24/T C-225	fernando.calle@upm.es	W - 14:00 - 15:00

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Estructura De Materiales Ii
- Electricidad Y Magnetismo
- Estructura De Materiales I
- Física De Materiales, Estadística Y Cuántica
- Propiedades De Materiales I
- Matemáticas Ii
- Properties Of Materials Ii

3.2. Other recommended learning outcomes

- Basic knowledge on modern Physics, electronic and optoelectronic devices included in the module "Introducción a la Electrónica"
- Written and spoken English

4. Skills and learning outcomes *

4.1. Skills to be learned

CE 2. - Saber modelizar el comportamiento (mecánico, electrónico, químico o biológico) de los materiales y su integración en componentes y dispositivos.

CE 5. - Capacitar para el aprendizaje autónomo de nuevos conocimientos y técnicas

CG 1 - Uso de la lengua inglesa

CG 11 - Responsabilidad y ética profesional

CG 2 - Capacidad de trabajo en equipo

CG 3 - Comunicación oral y escrita

4.2. Learning outcomes

RA109 - Knowledge on several Sustainable Development Goals, in particular GOAL 3: GOOD HEALTH AND WELL-BEING, GOAL 6: CLEAN WATER AND SANITATION, GOAL 7: AFFORDABLE AND CLEAN ENERGY, GOAL 9: INDUSTRY, INNOVATION AND INFRASTRUCTURE, and GOAL 17: PARTNERSHIPS FOR THE GOALS

RA29 - Entender, asimilar y manejar los conceptos básicos que describen el comportamiento electrónico, magnético, térmico y óptico de los materiales.

RA28 - Saber trabajar en equipo. Ejecutar el trabajo con responsabilidad y respeto a los demás.

RA27 - Utilizar con soltura la comunicación oral y escrita en lengua española e inglesa

RA30 - Conocer y saber relacionar dicho comportamiento con la estructura del material a nivel atómico, molecular y macroscópico.

RA31 - Conocer y saber calcular mediante las teorías más relevantes la respuesta electrónica, magnética, térmica y óptica de los materiales.

RA32 - Conocer los principios de funcionamiento y ser capaces de experimentar con dispositivos electrónicos elementales; transistores, láseres y otros sistemas

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The objective of this module is that students achieve a basic knowledge of the principles of nanoscience and nanotechnology, the properties of the main nanostructures, as well as the techniques used for their fabrication and characterisation. From the established microtechnology, the appearance and evolution of the main nanodevices will be reviewed, with a special focus on the fields of nanoelectronics, nanophotonics and nanobiotechnology. Finally, the module will include the application of such nanodevices in different areas relevant for structural and mainly functional material engineers, such as information and communication technologies, space, security, environment, domotics and nanomedicine.

From the content view, this module pursues opening the students mind to emerging ideas and technologies, and to provide a wide and multidisciplinary approach of the 21 c. science and technology, from the research towards the innovation. In the aptitude side, this module will increase their attitude and skill to think and relate contents; to research, elaborate and present information, and to work in a team.

5.2. Syllabus

1. Introduction and principles of nanotechnology

- 1.1. Emerging technologies
- 1.2. Market of nanotechnology
- 1.3. Precedents and history review
- 1.4. Scaling laws
- 1.5. Principles of quantum mechanics

2. Nanomaterials and nanostructures

- 2.1. Bonds and crystals
- 2.2. Inorganic semiconductors
- 2.3. Carbon nanostructures
- 2.4. Nanoparticles and composites
- 2.5. Organic compounds and biomaterials

3. Nanotechniques for fabrication and characterisation

- 3.1. Fabrication and manipulation techniques: deposition, lithography, self-assembly, molecular fabrication, nanomanipulation
- 3.2. Characterisation techniques: electrical, optical, structural (SEM and TEM, STM and AFM, nanoindentation)
- 3.3. Image treatment in nanotechnology
- 3.4. Visit to ISOM clean room

4. Nanodevices

- 4.1. Nanoelectronics
- 4.2. Nanooptoelectronics and nanophotonics
- 4.3. Nanobiotechnology

5. Workshop on Nanotechnology: Current applications and future perspectives

- 5.1. NT for building and structural materials
- 5.2. NT for energy and environment
- 5.3. NT for biotechnology and nanomedicine
- 5.4. NT for ICT

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	1: Introduction and principles of nanotechnology. Emerging technologies, market and history review Duration: 04:00 Lecture			Participation in class, presentation of readings, contribution to Moodle space Other assessment Progressive assessment Presential Duration: 00:00
2	1: Introduction and principles of nanotechnology (cont).Scaling laws. Principles of quantum mechanics Duration: 03:00 Lecture 1: Introduction and principles of nanotechnology (cont).Scaling laws. Principles of quantum mechanics Duration: 01:00 Problem-solving class			
3	2: Nanomaterials: Inorganic semiconductors Duration: 03:00 Lecture 2: Nanomaterials: inorganic semiconductors Duration: 01:00 Problem-solving class			
4	2: Nanomaterials: carbon nanostructures Duration: 03:00 Lecture 2: Nanomaterials: carbon nanostructures Duration: 01:00 Problem-solving class			
5	2: Nanomaterials: nanoparticles, biomaterials Duration: 03:00 Lecture 2: Nanomaterials: nanoparticles, biomaterials Duration: 01:00 Problem-solving class			

6	3: Nanotechniques: fabrication and manipulation Duration: 03:00 Lecture 3: Nanotechniques: fabrication and manipulation Duration: 01:00 Problem-solving class			
7	3: Nanotechniques: structural and electrical characterisation Duration: 02:00 Lecture Exam Duration: 02:00 Additional activities			Exam Part 1 Written test Progressive assessment Presential Duration: 00:00
8	3: Nanotechniques: structural and electrical characterisation Duration: 03:00 Lecture 3: Nanotechniques: fabrication and characterisation Duration: 01:00 Problem-solving class			
9	4: Nanodevices: Electronic properties Duration: 03:00 Lecture 4: Nanodevices: Electronic properties Duration: 01:00 Problem-solving class			
10	4: Electronic nanodevices Duration: 03:00 Lecture 4: Electronic nanodevices Duration: 01:00 Problem-solving class			
11	4: Nanodevices: optical properties Duration: 03:00 Lecture 4: Nanodevices: optical properties Duration: 01:00 Problem-solving class			
12	4: Optoelectronic nanodevices Duration: 03:00 Lecture 4: Optoelectronic nanodevices Duration: 01:00 Problem-solving class			

13	4: Nanobiotechnology. Nanosafety Duration: 02:00 Lecture Exam Duration: 02:00 Additional activities			Exam Part 2 Written test Progressive assessment Presential Duration: 00:00
14	5: Workshop: Applications of NT (Oral presentations) Duration: 04:00 Cooperative activities			
15	5: Workshop: Applications of NT (Oral presentations) Duration: 04:00 Cooperative activities			Oral presentation of teamwork. (This activity is compulsory, non-recoverable) Group presentation Progressive assessment and Global Examination Presential Duration: 00:00
16				
17				Final Exam Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Participation in class, presentation of readings, contribution to Moodle space	Other assessment	Face-to-face	00:00	%	0 / 10	CE 2. CE 5. CG 1 CG 2 CG 3 CG 11
7	Exam Part 1	Written test	Face-to-face	00:00	40%	3 / 10	CE 2. CE 5. CG 1 CG 3 CG 11
13	Exam Part 2	Written test	Face-to-face	00:00	40%	3 / 10	CE 2. CE 5. CG 1 CG 3 CG 11
15	Oral presentation of teamwork. (This activity is compulsory, non-recoverable)	Group presentation	Face-to-face	00:00	20%	4 / 10	CE 2. CE 5. CG 1 CG 2 CG 3 CG 11

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
15	Oral presentation of teamwork. (This activity is compulsory, non-recoverable)	Group presentation	Face-to-face	00:00	20%	4 / 10	CE 2. CE 5. CG 1 CG 2 CG 3 CG 11

17	Final Exam	Written test	Face-to-face	02:00	80%	4 / 10	CE 2. CE 5. CG 1 CG 11
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7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Written exam	Written test	Face-to-face	02:00	80%	5 / 10	CE 2. CE 5. CG 1 CG 3 CG 11
Oral presentation of teamwork (compulsory, non-recoverable activity)	Group presentation	Face-to-face	00:00	20%	5 / 10	CE 2. CE 5. CG 1 CG 2 CG 3 CG 11

7.2. Assessment criteria

The evaluation process will follow these guidelines:

CONTINUOUS EVALUATION: FINAL MARKS = 80 % exams + 20 % team work and presentation

Two partial exams will be done during the course and the end, around weeks 7 and 17. Each will count 40%. To pass, minimum grade in each exam should be >4/10.

The remaining 20% will be awarded from a team work and oral presentation given in the 12th to 14th weeks. This presentation is a compulsory, non-recoverable activity, due to its teamwork character. The weight will take into account the presentation itself (15%) and the participation during the workshop (5%).

Some recognition could also be given to those students who pass these exams from their individual contributions during classes, Moodle forums, etc.

GLOBAL EVALUATION

A global exam with two parts, each valued 40%, will be performed at the end of the module.

The remaining 20% will be awarded from the team work and oral presentation given in the 12th to 14th weeks.

EXTRAORDINARY CALL

An exam valued 80%

The remaining 20% will be awarded from the team work and oral presentation given in the 12th to 14th weeks.

In all cases, a complementary evaluation might take place to guarantee that original work has been performed.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Text book	Bibliography	- B. Rogers, S. Pennathur, J. Adams, Nanotechnology. Understanding small systems. CRC Press, 3rd ed. (2015).
Advanced book	Bibliography	- V.V. Mitin, V.A. Kochelap, M.A. Strocio, Introduction to nanoelectronics. Cambridge University Press (2012). - Rainer Waser (editor), Nanoelectronics and Information Technology. John Wiley & Sons (2012).

Advanced book 2	Bibliography	- Daniel Minoli, Nanotechnology Applications to Telecommunications and Networking, Wiley-Interscience (2005). - Bharat Bhushan (editor), Springer Handbook of Nanotechnology, 4th ed. Springer (2017).
Slides	Others	Disponibles en plataforma Moodle
Web links	Web resource	Selected web links regarding the state of the art, theory and lab training related to nanotechnology, nanoscience and their applications and impact in society
Classroom	Equipment	Assigned by the Program Coordinator
ISOM Labs	Equipment	A visit to the Clean Room ISOM facilities is planned

9. Other information

9.1. Other information about the subject

Assistance to classes and participation during the sessions is strongly recommended.

Also, inputs to the fora in the Moodle platform are suggested, either to contribute to discussion of doubts, or to propose debates on any topics related to Nanotechnology.